III. System Model and Problem Formulation

A. System Model

1) Network Scenario: We consider a general multi-beam LEO satellite communication system under NGSO–GSO coexistence. Since the LEO downlink and the GSO system may operate in the same frequency band, downlink transmissions from visible LEO satellites can generate co-channel interference at a GSO ground station. Hereafter, this ground station is referred to as the GS. The GS communicates with its target GSO satellite, and the direction from the GS to the target GSO satellite is used as the reference direction for EPFD evaluation. Among the LEO satellites visible from the GS, a near-inline NGSO–GSO geometry may occur when a satellite has a downlink beam whose footprint covers the GS and simultaneously passes close to the GS-to-GSO reference direction as observed from the GS, resulting in a higher interference risk.

The system is modeled in discrete time slots. Let $\mathcal{T}$ denote the set of time slots considered in this work, and let $t \in \mathcal{T}$ denote a specific time slot. For each time slot t, the satellite geometry, beam footprints, user coverage, and EPFD contribution may change due to LEO satellite motion. Therefore, the system state is evaluated on a time-slot basis.

Let $\mathcal{S}(t)$ denote the set of LEO satellites visible from the GS at time slot t, and let $\mathcal{B}_s$ denote the set of beams on satellite s. Each satellite employs a satellite-fixed multi-beam downlink layout [23], where each beam has a predefined pointing direction relative to the satellite and forms a corresponding ground footprint. For beam b on satellite s, the ground footprint at time slot t is denoted by $\mathcal{F}_{s,b}(t)$. If user u lies within $\mathcal{F}_{s,b}(t)$, beam b on satellite s is regarded as a feasible serving candidate for that user.

Because neighboring LEO satellites and their beams may have overlapping ground footprints, a terrestrial user may be covered by multiple beams from visible LEO satellites in the same time slot. This overlap provides multiple feasible serving beams for user association and resource coordination.

In this work, these feasible beams are assumed to operate with compatible downlink frequency and polarization settings. Therefore, changing the serving beam of a user can be modeled as a serving-beam reassociation among the beams that cover the user. This assumption is reasonable because predictable satellite motion, overlapping beam coverage, and compatible radio resources can support service continuity when a user is reassociated to another feasible beam [3]–[6].

2) User Association and Service Model: Let $\mathcal{U}(t)$ denote the set of terrestrial users considered at time slot t. Each user can be served by at most one beam in the same time slot. The user association is represented by a binary indicator $x_{u,s,b}(t)$, defined as

$$x_{u,s,b}(t) = \begin{cases} 1, & \text{if user } u \text{ is served by beam } b \text{ of satellite } s, \\ 0, & \text{otherwise,} \end{cases}$$

for all $u \in \mathcal{U}(t)$, $s \in \mathcal{S}(t)$, and $b \in \mathcal{B}_s$. A user can only be associated with a beam that covers its location. Therefore, if user u is not located within the footprint $\mathcal{F}_{s,b}(t)$, beam b of satellite s is not a feasible serving beam for this user.

For beam b of satellite s, let $N_{s,b}(t)$ denote the number of users served by this beam at time slot t. This value represents the beam loading, which reflects how many users share the available downlink resource of the same beam.

Each user u has a service demand. The demand represents the required downlink data rate of the user. In addition, different users may have different service priorities. Motivated by the priority concept in 3GPP QoS differentiation [15], each user u is assigned a priority class $\pi(u)$, and each priority class is associated with a priority weight $w_{\pi(u)}$. A higher priority weight indicates that the user has a higher importance in service recovery and resource allocation.

The user association state at time slot t is denoted by $\mathbf{X}(t)$, which contains all association indicators $x_{u,s,b}(t)$. The association decisions across all time slots are denoted by $\mathbf{X} = \{\mathbf{X}(t) \mid t \in \mathcal{T}\}$.

3) Beam Power Model: Each beam is assigned a transmit power in each time slot. Let $p_{s,b}(t)$ denote the transmit power of beam b on satellite s at time slot t. The beam power allocation state at time slot t is denoted by $\mathbf{P}(t)$, which contains all beam transmit powers $p_{s,b}(t)$. The beam power allocation decisions across all time slots are denoted by $\mathbf{P} = \{\mathbf{P}(t) \mid t \in \mathcal{T}\}$.

Under normal operation, each beam is assigned a predefined default transmit power. The default beam-power profile is configured according to beam geometry, antenna gain, and propagation loss. The purpose is to provide ground users in different beam footprints with similar received downlink power. This default setting represents the baseline power allocation before EPFD-aware power adjustment or beam shutdown is applied.

Each beam has a beam-level power upper bound, denoted by $P_{s,b}^{\max}$, and each satellite has a total transmit power budget, denoted by $P_{s,tot}^{\max}$. During EPFD-aware operation, the transmit power of a beam may be adjusted from its default value. A beam is regarded as shut down when its transmit power is set to zero. The beam-level and satellite-level power limits are included later in the problem formulation as power constraints.

4) SINR Model: The downlink SINR of a user is modeled using a standard wireless and satellite communication formulation, where the received desired signal, LEO co-channel interference, and receiver noise are considered. If user u is served by beam b of satellite s at time slot t, its downlink SINR is given by

$$\mathrm{SINR}_u(t) = \frac{S_u(t)}{I_u^{\mathrm{intra}}(t) + I_u^{\mathrm{inter}}(t) + N_u(t)}, \tag{1}$$

where $S_u(t)$ is the desired signal power, $I_u^{\mathrm{intra}}(t)$ is the LEO intra-satellite co-channel interference, $I_u^{\mathrm{inter}}(t)$ is the LEO inter-satellite co-channel interference, and $N_u(t)$ is the receiver noise power.

Desired Signal Power: The desired signal power follows the standard satellite link-budget relation. For user u served by beam b of satellite s, the received desired signal power is given by

$$S_u(t) = \frac{p_{s,b}(t)\, G_{s,b\to u}^{\mathrm{tx}}(t)\, G_u^{\mathrm{rx}}}{L_{s,u}(t)},$$

where $p_{s,b}(t)$ is the transmit power of beam b on satellite s, $G_{s,b\to u}^{\mathrm{tx}}(t)$ is the transmit antenna gain of beam b toward user

u, G_u^{rx} is the user terminal receive gain, and $L_{s,u}(t)$ is the total propagation loss between satellite s and user u.

The total propagation loss includes free-space path loss and atmospheric gaseous loss:

$$L_{s,u}(t) = L_{s,u}^{\mathrm{fs}}(t)\, L_{s,u}^{\mathrm{gas}}(t),$$

where $L_{s,u}^{\mathrm{fs}}(t)$ is the free-space path loss [13] and $L_{s,u}^{\mathrm{gas}}(t)$ is the atmospheric gaseous loss [14]. The free-space path loss is given by

$$L_{s,u}^{\mathrm{fs}}(t) = \left(\frac{4\pi d_{s,u}(t)}{\lambda}\right)^2,$$

where $d_{s,u}(t)$ is the distance between satellite s and user u, and λ is the carrier wavelength.

LEO Intra-Satellite Interference: The LEO intra-satellite interference comes from co-channel beams on the same satellite [21]. Let $\mathcal{B}_s^{\mathrm{co}}(b)$ denote the set of beams on satellite s that use the same downlink frequency resource as the serving beam b, excluding beam b itself. The intra-satellite interference received by user u is given by

$$I_u^{\mathrm{intra}}(t) = \sum_{b' \in \mathcal{B}_s^{\mathrm{co}}(b)} \frac{p_{s,b'}(t)\, G_{s,b' \to u}^{\mathrm{tx}}(t)\, G_u^{\mathrm{rx}}}{L_{s,u}(t)}.$$

LEO Inter-Satellite Interference: The LEO inter-satellite interference comes from co-channel beams on other visible LEO satellites. For another visible satellite s', let $\mathcal{B}_{s'}^{\mathrm{co}}(b)$ denote the set of beams on satellite s' that use the same downlink frequency resource as the serving beam b. The inter-satellite interference received by user u is given by

$$I_u^{\mathrm{inter}}(t) = \sum_{\substack{s' \in \mathcal{S}(t) \\ s' \neq s}} \sum_{b' \in \mathcal{B}_{s'}^{\mathrm{co}}(b)} \frac{p_{s',b'}(t)\, G_{s',b' \to u}^{\mathrm{tx}}(t)\, G_u^{\mathrm{rx}}}{L_{s',u}(t)},$$

where $G_{s',b' \to u}^{\mathrm{tx}}(t)$ is the transmit antenna gain of the interfering beam toward user u, and $L_{s',u}(t)$ is the total propagation loss between satellite s' and user u.

With this SINR model, the user SINR changes when the serving beam, beam transmit power, propagation loss, or LEO co-channel interference changes.

5) Capacity and User Satisfaction Model: After obtaining the user SINR, the achievable downlink capacity is modeled by the Shannon capacity formula [22], which is commonly used in wireless and satellite communication systems. Let B denote the downlink bandwidth of a beam. If user u is served by beam b of satellite s at time slot t, the achievable downlink capacity of user u is given by

$$C_u(t) = B \log_2\!\big(1 + \mathrm{SINR}_u(t)\big).$$

When multiple users are served by the same beam, they share the downlink resource of that beam. In this work, users served by the same beam are assumed to equally share the beam resource in each time slot. The actual service rate of user u is therefore

$$R_u(t) = \frac{C_u(t)}{N_{s,b}(t)}, \tag{2}$$

where $N_{s,b}(t)$ represents the beam loading of beam b on satellite s at time slot t. This means that users served by the same beam are allocated the same fraction of the available beam resource.

Let D_u denote the service demand of user u. Following the demand satisfaction concept used in [18], the satisfaction of user u is calculated by comparing the actual service rate with the user demand and is defined as

$$s_u(t) = \min\left(\frac{R_u(t)}{D_u}, 1\right), \tag{3}$$

where $s_u(t) \in [0,1]$. If $s_u(t) = 1$, the demand of user u is fully met; if $0 < s_u(t) < 1$, user u receives partial service; if $s_u(t) = 0$, user u receives no effective service in the time slot. This satisfaction metric represents the fraction of user demand that is satisfied in each time slot.

6) EPFD Model: The EPFD calculation follows the ITU Radio Regulations and ITU-R S.1503 framework for NGSO–GSO coexistence [9], [10]. In this work, the EPFD at the GS is calculated by first computing the contribution of each active beam from visible LEO satellites. For each active beam, the contribution depends on its transmit power, transmit antenna gain toward the GS, propagation distance, and the receive antenna gain of the GS. The contributions of all active beams from all visible LEO satellites are then summed in the linear domain to obtain the aggregate EPFD at the GS. This calculation considers each active beam separately because different beams on the same LEO satellite may have different pointing directions, antenna gains, and transmit powers toward the GS.

For EPFD calculation, let $P_{s,b}^{\mathrm{ref}}(t)$ denote the radio-frequency (RF) transmit power of beam b on satellite s within the ITU reference bandwidth B_{ref} specified for the applicable EPFD limit [11], [12]. If the beam transmit power $p_{s,b}(t)$ is defined over a wider beam bandwidth, it is converted to the corresponding RF power within B_{ref} before applying the EPFD formula. The EPFD contribution of beam b on satellite s toward the GS is given by

$$\mathrm{EPFD}_{s,b}(t) = \frac{P_{s,b}^{\mathrm{ref}}(t)\, G_{s,b\to g}^{\mathrm{tx}}(t)}{4\pi d_{s,g}^2(t)} \cdot \frac{G_g^{\mathrm{rx}}(\theta_{g\to s}(t))}{G_{g,\max}^{\mathrm{rx}}},$$

where $G_{s,b\to g}^{\mathrm{tx}}(t)$ is the transmit antenna gain of beam b on satellite s in the direction of the GS, $d_{s,g}(t)$ is the distance between satellite s and the GS, $G_g^{\mathrm{rx}}(\theta_{g\to s}(t))$ is the receive antenna gain of the reference GS antenna in the direction of satellite s, $G_{g,\max}^{\mathrm{rx}}$ is the maximum receive antenna gain of the reference GS antenna, and $\theta_{g\to s}(t)$ is the off-axis angle between the GS antenna boresight and the direction of satellite s.

The transmit antenna gain $G_{s,b\to g}^{\mathrm{tx}}(t)$ is evaluated from the beam antenna pattern according to the off-axis angle between the boresight of beam b and the direction of the GS. Since each beam has its own pointing direction, the EPFD contribution is computed separately for each beam. If a beam is shut down, its transmit power is zero and its EPFD contribution is also zero.

The aggregate EPFD at the GS is obtained by summing the EPFD contributions of all active beams of all visible LEO satellites:

$$\mathrm{EPFD}_{\mathrm{agg}}(t) = \sum_{s\in\mathcal{S}(t)} \sum_{b\in\mathcal{B}_s} \mathrm{EPFD}_{s,b}(t).$$

The resulting aggregate EPFD is compared with the applicable EPFD limit in the problem formulation.

B. Problem Formulation

Based on the system model above, we formulate the user association and beam power allocation problem under user association, beam power, satellite power budget, and EPFD constraints. The decision variables are the user association sequence $\mathbf{X}$ and the beam power allocation sequence $\mathbf{P}$. Specifically, $\mathbf{X}(t) = \{x_{u,s,b}(t)\}$ denotes the user association state at time slot t, and $\mathbf{P}(t) = \{p_{s,b}(t)\}$ denotes the beam power allocation state at time slot t.

Objective function:

$$\max_{\mathbf{X},\mathbf{P}} \sum_{t\in\mathcal{T}} \sum_{u\in\mathcal{U}(t)} w_{\pi(u)}\, s_u(t). \tag{4}$$

Constraints:

$$\sum_{s\in\mathcal{S}(t)} \sum_{b\in\mathcal{B}_s} x_{u,s,b}(t) \le 1. \tag{5}$$

$$x_{u,s,b}(t) \in \{0,1\}. \tag{6}$$

$$0 \le p_{s,b}(t) \le P_{s,b}^{\max}. \tag{7}$$

$$\sum_{b\in\mathcal{B}_s} p_{s,b}(t) \le P_{s,tot}^{\max}. \tag{8}$$

$$\mathrm{EPFD}_{\mathrm{agg}}(t) \le \mathrm{EPFD}_{\mathrm{lim}}. \tag{9}$$

where (5) holds for all $u \in \mathcal{U}(t)$ and $t \in \mathcal{T}$; (6) holds for all $u \in \mathcal{U}(t)$, $s \in \mathcal{S}(t)$, $b \in \mathcal{B}_s$, and $t \in \mathcal{T}$; (7) holds for all $s \in \mathcal{S}(t)$, $b \in \mathcal{B}_s$, and $t \in \mathcal{T}$; (8) holds for all $s \in \mathcal{S}(t)$ and $t \in \mathcal{T}$; and (9) holds for all $t \in \mathcal{T}$.

Constraint (5) ensures that each user is served by at most one beam in each time slot. As described in the user association model, only beams that cover the user are considered feasible serving beams.

Constraint (6) defines the binary user association variable.

Constraint (7) limits the transmit power of each beam and includes the possibility of beam shutdown when $p_{s,b}(t) = 0$.

Constraint (8) enforces the total transmit power budget of each satellite.

Constraint (9) ensures that the aggregate EPFD at the GS does not exceed the applicable EPFD limit.

The objective in (4) maximizes the weighted satisfaction of users. The weight $w_{\pi(u)}$ reflects the service priority of user u, while $s_u(t)$ represents the fraction of the user demand that is satisfied in time slot t. Therefore, higher-priority users can receive greater importance in the objective function.

The association and power allocation decisions are strongly coupled. User association affects beam loading, downlink rate, and user satisfaction, while beam power affects both downlink SINR and the aggregate EPFD at the GS. In near-inline NGSO–GSO geometries, shutting down beams can lower the aggregate EPFD, but it may also reduce the service rates of users originally served by the beams that are shut down. Therefore, the problem becomes a mixed-integer nonlinear optimization problem. Solving the global optimum over all time slots is computationally difficult, so the next chapter develops EPFD-Aware Beam Reassociation for Service Recovery to obtain feasible association and power allocation decisions under the EPFD constraint.

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